Day for Night filter

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***Abstract*—This paper presents a Day-for-Night (DFN) image processing pipeline that transforms daylight photographs and video frames into visually coherent nighttime scenes. The proposed MATLAB implementation operates mainly in linear RGB space and employs a Retinex-inspired illumination-reflectance decomposition for local relighting. This approach is combined with gamma-based darkening, cool moonlight color adaptation, linear photometric saturation reduction, soft sky-specific masking, visual-acuity blur, and synthetic low-light noise. In addition, EXIF metadata is used to guide exposure reduction, vignetting, and noise strength, making the transformation physically grounded and scene-specific. Experiments on high-quality images extracted from the RAISE dataset, as well as on video frames, show that the proposed pipeline effectively reduces overall luminance, shifts the color appearance toward cooler nighttime tones, and preserves high-frequency structural details better than a purely global darkening or tone-mapping approach. Histogram analysis confirms the expected shift of pixel intensities toward darker ranges after processing.**

***Index Terms*—day-for-night, tone mapping, image relighting, color grading, low-light simulation, non-photorealistic rendering.**

I. INTRODUCTION

The Day-For-Night (DFN) technique, widely used in Hollywood cinematography, consists of capturing footage in daylight conditions and applying post-processing to simulate a night scene [1]. This approach allows filmmakers to shoot at high quality (low ISO, good dynamic range) while still achieving a nighttime aesthetic in post-production.

From an image processing perspective, the night appearance is characterised by three main visual cues: (i) a strong reduction in overall luminance, (ii) a cool blue tint due to the dominance of moonlight and skylight [2], and (iii) increased noise and grain at higher ISO equivalent settings.

This paper proposes a MATLAB implementation of a metadata-guided Day-for-Night (DFN) pipeline. The main processing stages are:

1) conversion of the input image from display-referred sRGB to linear RGB;
2) extraction of a baseline photopic luminance map;
3) Retinex-inspired illumination-reflectance decomposition and local relighting to preserve micro-contrast;
4) cool moonlight color adaptation through RGB channel scaling;
5) linear photometric saturation reduction to imitate weaker scotopic color perception;
6) soft sky-region estimation and dedicated alpha-blended sky darkening;
7) visual-acuity blur, detail sharpening, and Gaussian noise injection for still images;
8) Poisson shot-noise modeling for video frames;
9) metadata-based calibration of exposure, vignetting, and noise parameters;
10) extension of the pipeline to video frames with attention to temporal consistency.

The remainder of this paper is organized as follows. Section II describes the image selection and preprocessing procedure. Section III presents the proposed Day-for-Night algorithm. Section IV discusses the experimental results, including visual comparisons, video-frame results, and histogram analysis. Section V concludes the paper and outlines possible future improvements.

II. IMAGE DATA PREPROCESSING

To evaluate the Day-for-Night pipeline on high-quality, uncompressed data, source images were extracted from the RAISE dataset [5]. RAISE provides camera-native RAW (NEF) and high-fidelity TIFF images, ensuring that the initial daylight images are free from lossy compression artifacts (such as JPEG blocking) that could interfere with tone mapping and synthetic noise injection.

A custom MATLAB utility parses the dataset CSV metadata and extracts images according to their EXIF profiles. Since the original RAISE images may be stored with different numerical formats, the selected TIFF files are standardized before being processed by the DFN pipeline. In the current implementation, 16-bit integer images are scaled from the range $[0, 65535]$ to $[0, 255]$ and stored as 8-bit images, while floating-point images are clamped to the interval $[0, 1]$. The resulting daylight images are exported as losslessly compressed TIFF files using LZW compression. This provides a consistent input format for the linear RGB conversion described in Section III-A.

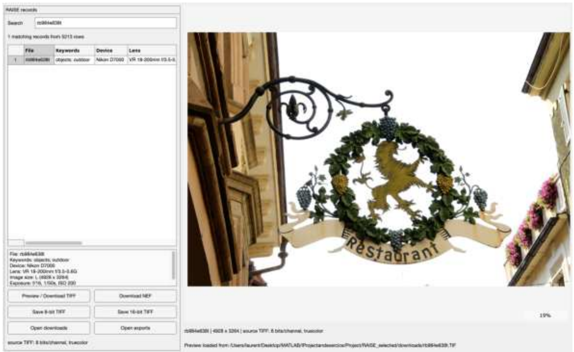


Fig. 1: The custom MATLAB interface developed to browse the RAISE dataset, displaying image metadata (EXIF) and facilitating the selection and extraction of uncompressed source files.

III. Description of the Algorithm

The proposed Day-for-Night (DFN) pipeline models key aspects of human scotopic vision and nighttime image formation. The method operates in linear RGB space and applies a sequence of luminance, chromatic, and semantic transformations to convert daytime photographs into visually coherent night scenes.

A. sRGB to Linear Conversion

The input image is first converted from the standard sRGB color space to linear RGB. Since sRGB values are gamma–encoded according to the IEC 61966–2–1 transfer function, pixel intensities do not correspond linearly to physical scene radiance. Because subsequent operations—such as scotopic luminance computation, Retinex-based relighting, color temperature adjustment, and noise modelling—require a physically meaningful representation of light, the nonlinear sRGB encoding must be inverted. The linearization step applies the inverse sRGB electro–optical transfer function to each channel:

$$I_{\text{lin}} = \begin{cases} \dfrac{I_{\text{srgb}}}{12.92}, & I_{\text{srgb}} \le 0.04045, \\ \left(\dfrac{I_{\text{srgb}} + 0.055}{1.055}\right)^{2.4}, & I_{\text{srgb}} > 0.04045. \end{cases} \tag{1}$$

This transformation ensures that pixel intensities become proportional to actual luminance, providing a consistent foundation for all subsequent stages of the Day-for-Night pipeline.

B. Luminance Computation

After linearization, the scene's base luminance is extracted. While human scotopic vision shifts sensitivity toward blue-green wavelengths, the proposed pipeline primarily structures its illumination processing around standard photopic (cone-driven) luminance to preserve baseline scene contrast before simulated darkening. The luminance Y is obtained through the standard Rec. 709 weighted sum of the linear RGB channels:

$$Y = 0.2126R + 0.7152G + 0.0722B. \tag{2}$$

This luminance map serves as the fundamental basis for the subsequent illumination-reflectance decomposition.

C. Retinex-Based Illumination and Relighting

Compared with global tone-mapping operators such as Reinhard's photographic tone reproduction [6], the Retinex-inspired decomposition is better suited to the Day-for-Night task because it modifies the estimated illumination component while preserving reflectance details such as edges and textures. The proposed pipeline employs a local Retinex-inspired spatial decomposition from [6]. This isolates the scene's lighting from its intrinsic textures, allowing for deep exposure reduction without losing high-frequency details.

1) Illumination-Reflectance Decomposition: The scene's low-frequency illumination map L_{ill} is estimated by applying a Gaussian low-pass filter to the luminance map Y:

$$L_{\text{ill}} = \mathcal{G}_{\sigma}(Y) \tag{3}$$

where $\mathcal{G}_{\sigma}$ denotes a Gaussian filter with standard deviation σ. The intrinsic scene reflectance R_{ref}, which contains local details and textures, is then extracted by dividing the linear RGB image by the illumination map:

$$R_{\text{ref}} = \min\left(\frac{RGB_{\text{linear}}}{L_{\text{ill}}}, 2.4\right) \tag{4}$$

2) Target Night Illumination: The daylight illumination is mathematically transformed into a target night illumination field L_{target} by applying an exposure reduction factor E, gamma darkening γ, a contrast multiplier C, and a black floor offset B:

$$L_{\text{target}} = C \times (E \cdot L_{\text{ill}}^{\gamma}) + B \tag{5}$$

Optical vignetting $V(x, y)$ is also multiplied into this field to simulate lens fall-off.

3) Relighting: The final step of this stage reconstructs the image by multiplying the original reflectance details by the newly generated night illumination field:

$$RGB_{\text{out}} = R_{\text{ref}} \times L_{\text{target}} \tag{6}$$

This spatial approach drastically darkens the overall scene while strictly preserving high-frequency micro-contrast and local edges.

D. Moonlight Color Adaptation

Following luminance compression, a chromatic adaptation step is applied to reproduce the characteristic blue tint of nocturnal illumination. Although moonlight is reflected sunlight, its perceived color temperature is significantly cooler due to the combined influence of atmospheric scattering and the increased sensitivity of rod photoreceptors to short wavelengths. To approximate this effect, the RGB channels are scaled using a simple linear model that attenuates long wavelengths while enhancing the blue component:

$$[R',\, G',\, B'] = [0.62R,\ 0.74G,\ 1.12B]. \tag{7}$$

This transformation shifts the image toward a cooler chromatic appearance while remaining computationally lightweight. The operation also implicitly reflects the Purkinje effect, whereby red hues become less perceptible under low-light conditions. The resulting color adaptation provides the foundational moonlit aesthetic that characterizes nighttime scenes.

E. Saturation Reduction

Following colour temperature adjustment, colour saturation is reduced to simulate the loss of colour perception under low-light conditions. Since rod photoreceptors are largely insensitive to colour, objects appear less saturated at night.

Rather than translating the image into non-linear color spaces (such as HSV), saturation reduction is performed photometrically in linear space. The desaturated image is computed via linear interpolation between the relit RGB tensor and its grayscale luminance equivalent Y_{relit}:

$$RGB_{\text{desat}} = Y_{\text{relit}} + k \times (RGB_{\text{relit}} - Y_{\text{relit}}) \tag{8}$$

where $0 \leq k \leq 1$ is the saturation scaling factor ($k = 0.30$ in this implementation). This linear channel mixing preserves the strict photometric energy established during the relighting phase while muting vibrant daytime colours.

F. Sky Processing

To enhance the realism of the Day-for-Night conversion, the sky region is identified and processed separately. In real night scenes, the sky typically appears significantly darker than its daytime counterpart.

1) Soft Sky Mask Estimation: Instead of relying on hard binary thresholds and morphological operators, the proposed method estimates a continuous (soft) alpha mask $M(x, y) \in [0, 1]$. This mask is derived by identifying regions that satisfy specific hue boundaries (blue dominance), minimum saturation, and sufficient brightness. To enforce a spatial prior—since skies reliably appear at the top of an outdoor image—the mask is scaled by a vertical linear gradient weight:

$$W(y) = 1.0 - 0.8 \left(\frac{y}{H_{\text{img}}} \right) \tag{9}$$

Finally, the mask is smoothed using a Gaussian filter to ensure soft anti-aliased transitions along the horizon and around foreground objects, preventing harsh masking artifacts.

2) Sky Darkening: Using the soft mask, pixels belonging to the sky region are selectively darkened using an alpha-blended attenuation factor β_{sky}:

$$RGB_{\text{final}} = RGB_{\text{desat}} \times (1 - \beta_{\text{sky}} \cdot M(x, y)) \tag{10}$$

This operation smoothly pushes the sky's luminance correctly into the nighttime range without introducing visible seams.

G. Final Polishing

1) Acuity Blur and Detail Sharpening: Human visual acuity drops significantly in scotopic conditions. To model this, a mild visual-acuity blur is applied using a linear blend between the image and a Gaussian-blurred version of itself. To offset the synthetic softness and mimic the perceptual edge-contrast of a camera lens operating in low light, an unsharp masking operation is subsequently applied. This combination of a low-pass Gaussian filter and high-frequency edge sharpening realistically simulates the optical properties of night photography better than a standalone edge-preserving filter.

2) Noise Injection: The final stage of the still-image Day-for-Night pipeline introduces synthetic noise to reproduce the grainy appearance often observed in low-light photography. In the image pipeline, this effect is modeled using additive Gaussian noise. This choice provides a simple and controllable approximation of sensor read noise and post-amplification grain after the image has been darkened.

The noisy image is computed as

$$I_{\text{noisy}} = I + n, \tag{11}$$

where I is the processed image and n is a zero-mean Gaussian random variable:

$$n \sim \mathcal{N}(0, \sigma^2_{\text{night}}). \tag{12}$$

The standard deviation σ_{night} controls the visibility of the grain. In this work, it is adjusted according to the simulated ISO increase, so that images requiring a stronger low-light amplification receive a higher noise level. The result is then clipped to the valid intensity range to avoid invalid pixel values.

This Gaussian noise model is used for still images because it is visually effective, stable, and easy to control. For the video extension, a Poisson noise model is used instead, as described in Section III-I, in order to better approximate signal-dependent shot noise across frames.

H. Physically-Motivated Parameter Calibration

To avoid arbitrary hand-tuning, the pipeline extracts EXIF metadata to establish physically motivated parameters for the Day-for-Night conversion. The required luminance reduction is derived by calculating the daylight scene's Exposure Value (EV_{100}) using the camera's f-number (f) and shutter speed (t):

$$EV_{100} = \log_2 \left(\frac{f^2}{t} \right)$$

This allows us to compute the exact scaling factor required to reach a target scotopic luminance. Furthermore, low-light sensor grain is accurately simulated by scaling the camera's base read noise (σ_{base}) against the simulated ISO shift:

$$\sigma_{night} = \sigma_{base} \sqrt{\frac{ISO_{night}}{ISO_{day}}}$$

Finally, we replace standard Gaussian darkening with a physically accurate optical vignetting model. Using the lens focal length and sensor diagonal, light fall-off is applied proportionally to $\cos^4(\theta)$, where θ is the angle from the optical axis.

I. Temporal Consistency and Video Processing

Extending the Day-for-Night pipeline to video introduces challenges regarding temporal consistency, HDR color spaces, and computational efficiency. To process the high-dynamic-range iPhone footage (10-bit HLG / Dolby Vision),the pipeline first applies the inverse ARIB STD-B67 opto-electronic transferfunction (OETF) to linearize the signal before nighttime relighting and luminance compression:

$$E_{linear} = \begin{cases} \frac{E^2}{3}, & 0 \leq E \leq 0.5 \\ \frac{\exp((E-c)/a)+b}{12}, & 0.5 < E \leq 1 \end{cases} \tag{13}$$

where $a = 0.1788$, $b = 0.2846$, and $c = 0.5599$. This ensures that super-bright HDR pixels are safely compressed into standard display-referred space prior to nighttime darkening.

Furthermore, synthetic shot noise is introduced to simulate a low-light sensor. In real night-time photography, the reduced number of incoming photons causes the sensor to exhibit noticeable shot noise, which appears as random grain. Shot noise is signal dependent: brighter regions have larger absolute photon-count variation, while darker regions often appear noisier relative to their signal because their signal-to-noise ratio is lower. It is mathematically modeled using a Poisson distribution:

$$I_{noisy} = \frac{\mathcal{P}(I_{clean} \cdot N_{peak})}{N_{peak}} \tag{14}$$

where $\mathcal{P}(\lambda)$ represents a random sample from the Poisson distribution and N_{peak} defines the maximum photon count, dynamically set (e.g., to 140) based on the video metadata. To control the visual strength of the effect, this noisy result is then blended with the smoothed image:

$$I_{out} = (1 - \alpha)I + \alpha I_{noisy} \tag{15}$$

Here, I is the processed image, I_{noisy} is the Poisson-corrupted version, and α determines how visible the noise becomes. However, applying independent noise models to successive frames introduces severe temporal flickering. To prevent this, spatial operations like bilateral filtering are bypassed, and the random seed for the Poisson noise generation is deterministically linked to the frame index.

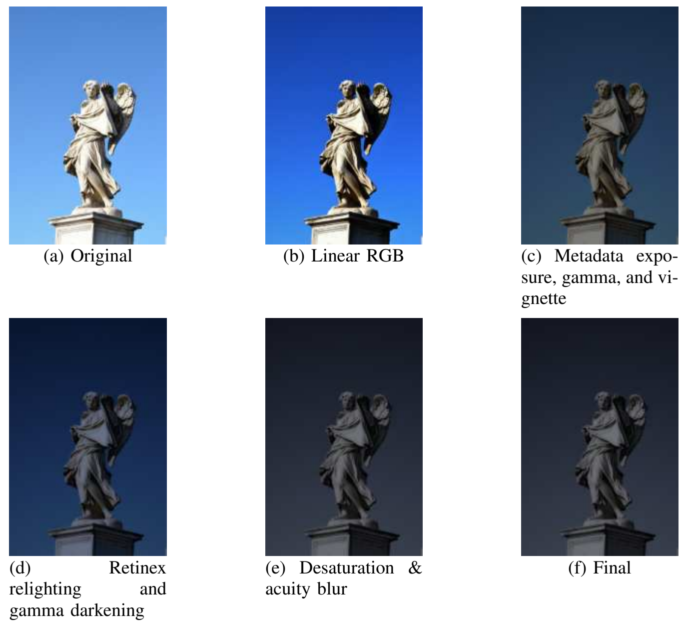
Finally, the pipeline utilizes the video's EXIF timestamp and GPS metadata to dynamically calculate the solar zenith angle θ_z:

$$\cos(\theta_z) = \sin(\phi)\sin(\delta) + \cos(\phi)\cos(\delta)\cos(h) \tag{16}$$

where ϕ is the capture latitude, δ is the solar declination, and h is the hour angle. The resulting solar elevation $(90° - \theta_z)$ dictates the initial daytime scene brightness, allowing the luminance exposure drop to scale according to actual physical lighting conditions rather than arbitrary artistic tuning.

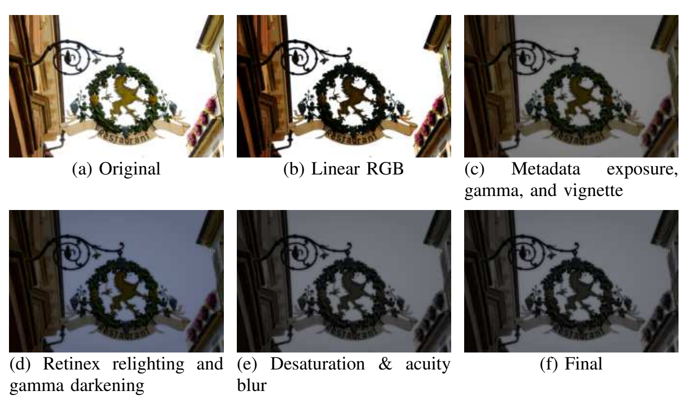
IV. Experimental Results

A. Result for Each Processing Step of the 2 Images



(a) Original (b) Linear RGB (c) Metadata exposure, gamma, and vignette (d) Retinex relighting and gamma darkening (e) Desaturation & acuity blur (f) Final

Fig. 2: Step-by-step processing of the Day-for-Night pipeline for the 1st image.



(a) Original (b) Linear RGB (c) Metadata exposure, gamma, and vignette (d) Retinex relighting and gamma darkening (e) Desaturation & acuity blur (f) Final

Fig. 3: Step-by-step processing of the Day-for-Night for the 2nd image.

B. Result for 3 frames of the Video processing

Fig. 4: Three consecutive frames from the video pipeline, showing temporal consistency of the Day-for-Night effect.

C. Overall Result

The Day-for-Night pipeline effectively converts the original daylight image into a convincing nighttime scene. Starting from the linear RGB conversion (Figure 2b), each step gradually shifts the image toward lower brightness and cooler color tones. Retinex-based relighting and gamma darkening reduce the overall luminance (Figures 2d and 3d), while the cool color adjustment introduces the characteristic bluish tint of night illumination. Saturation reduction further mutes warm colors (Figure 2e), and sky processing darkens the background more strongly than the foreground, producing a realistic night sky gradient (Figure 2d). Edge-aware smoothing softens fine textures without blurring important contours, and the final noise addition introduces subtle grain typical of low-light photography. Crucially, leveraging EXIF metadata to dynamically calibrate these parameters—such as the targeted luminance drop and synthetic noise scaling—ensures the transformation remains physically grounded. Together, these steps produce a coherent night appearance while preserving the structure and detail of the original scenes (Figures 2f, 3f and 4).

a) Parameter Choice and Metadata Calibration: To maintain a consistent nighttime aesthetic, both images share several core visual settings. A gamma of $\gamma = 1.78$ provides necessary overall darkening, while moonlight color multipliers ([0.62,0.74,1.12]) and a saturation factor of 0.30 simulate the cool, muted tones of human scotopic vision. Additionally, both scenes apply a sky-darkening strength of 0.55, alongside a mild visual-acuity blur ($\sigma = 1.0$) and a slight sharpening pass to replicate reduced low-light focus. Crucially, rather than relying purely on manual artistic tuning, the exposure, vignetting, and noise are dynamically calibrated using each image's unique EXIF metadata:

- **Exposure:** The baseline darkening factor is automatically calculated from the original camera settings (f-stop, shutter speed, and ISO) to hit a uniform target night illumination. This resulted in an exposure factor of 0.2585 for the statue and 0.2891 for the restaurant sign.
- **Vignetting:** When the physical focal length is known (as with the statue), an accurate $cos(4\theta)$ light fall-off model is applied. If this metadata is missing (as with the sign), the pipeline safely defaults to a standard radial vignette.
- **Noise:** Simulated sensor grain scales proportionally to the required ISO jump. Because the restaurant sign required a much larger simulated shift (ISO 200 to ISO 6400) than the statue (ISO 800 to ISO 6400), its resulting calculated Gaussian noise is accurately stronger.

Summary: The two images use the same core visual pipeline, but the metadata directly dictates the exposure, noise level, and vignetting model, which makes the method physically scene-specific rather than purely hand-tuned.

D. Histogram Analysis

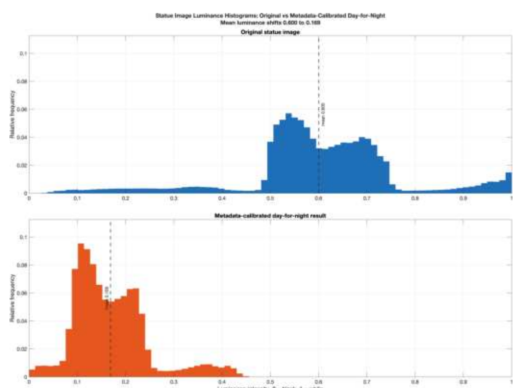


Fig. 5: Histograms comparing the original daylight image and the Day-for-Night result for the statue scene.

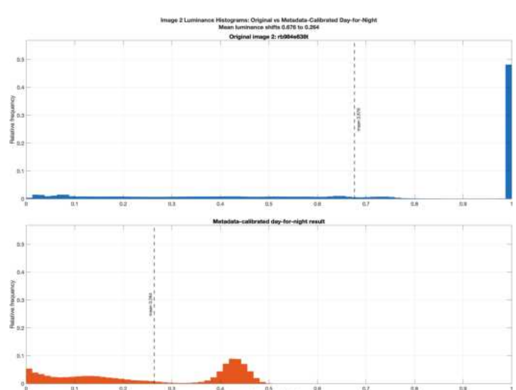


Fig. 6: Histograms comparing the original daylight image and the Day-for-Night result for the restaurant sign scene.

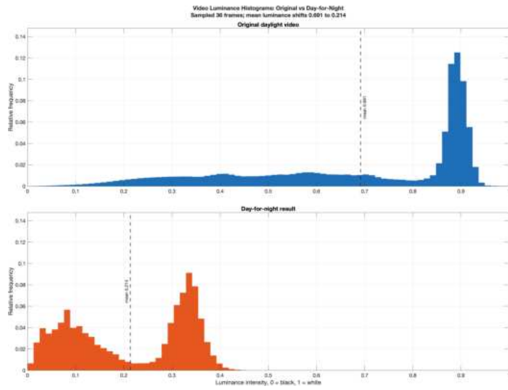


Fig. 7: Histograms comparing an original daylight frame and the Day-for-Night result from the video sequence.

The histogram comparison shows a clear shift in pixel intensities after applying the Day-for-Night pipeline. The statue image has most values in the mid-to-high intensity range, reflecting bright daylight. The restaurant image also has stacked intensities in the high range. In contrast, the night-image histogram is concentrated toward lower intensities, confirming that the Retinex-based relighting and darkening steps effectively reduced overall brightness. The narrower spread also indicates a compressed dynamic range, which is typical of low-light scenes. Overall, the histogram validates that the pipeline successfully transforms the luminance distribution to match nighttime conditions.

V. Conclusion

This project presented a multi-stage Day-for-Night pipeline that transforms daylight photographs and video frames into visually coherent night-time scenes. By combining linear-RGB processing, Retinex-inspired local relighting, cool-color adjustment, linear photometric saturation control, soft alpha-blended sky darkening, visual-acuity blur with detail sharpening, and synthetic low-light noise, the method reproduces the key visual cues of nighttime imaging while preserving the high-frequency structure of the original scene. In the still-image pipeline, Gaussian noise is used to simulate controllable low-light grain, while the video extension uses Poisson noise to approximate signal-dependent shot noise with improved temporal consistency. The results demonstrate that a metadata-guided, step-wise approach can achieve a convincing night-time appearance without requiring complex learning-based models.

Future directions include:

- Semantic masking to preserve bright artificial light sources (lamps, windows)
- Deep-learning refinement using a GAN-based night-style transfer
- Integrating wavelet decomposition to enable spatially correlated noise shaping, localized tone mapping, and temporal wavelet-based processing for video.

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References

[1] X. Xing, W. Dong, X. Zhang, and J. C. Paul, "Spectrally-based single image relighting," in *Proc. Int. Conf. Technologies for E-Learning and Digital Entertainment*, pp. 509–517, Springer, 2010.

[2] G. J. Ward, "Measuring and modeling anisotropic reflection," in *ACM SIGGRAPH Computer Graphics*, vol. 26, no. 2, pp. 265–272, 1992.

[3] A. Levin, D. Lischinski, and Y. Weiss, "Colorization using optimization," in *ACM Trans. Graph.*, vol. 23, no. 3, pp. 689–694, 2004.

[4] K. He, J. Sun, and X. Tang, "Guided image filtering," in *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 6, pp. 1397–1409, 2013.

[5] D.-T. Dang-Nguyen, C. Pasquini, V. Conotter, and G. Boato, "RAISE – A Raw Images Dataset for Digital Image Forensics," in *ACM Multimedia Systems*, Portland, Oregon, March 18–20, 2015.

[6] E. H. Land and J. J. McCann, "Lightness and Retinex theory," *Journal of the Optical Society of America*, vol. 61, no. 1, pp. 1–11, 1971.

[7] E. Reinhard, M. Stark, P. Shirley, and J. Ferwerda, "Photographic tone reproduction for digital images," *ACM Transactions on Graphics*, vol. 21, no. 3, pp. 267–276, 2002.

Appendix: Group Contribution

- **Béguelin Laurent:** Designed and implemented the image selector, integrated metadata-based parameter calibration to improve the filter, and applied the method to video using OETF and Poisson noise. Conducted research on the theory behind Day-for-Night filters and wrote Sections I, II, and IV.
- **Canh Trieu:** Implemented the main processing stages, including contrast adjustment, noise generation, desaturation, and blur. Built the evaluation framework, produced all figures, and wrote Sections III and V.